

KUOPIO, Finland

WMO: 029170

Lat: 63.0083N

Lon: 27.7994E

Elev: 99

StdP: 100.15

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.3	-24.0	-30.1	0.2	-27.1	-26.5	0.3	-24.0	10.3	-2.2	9.3	-2.0	1.0	310	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	26.6	18.3	24.8	17.4	23.0	16.6	19.7	24.1	18.6	22.8	17.6	21.5	3.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.2	21.5	17.1	12.3	20.6	16.0	11.5	19.8	56.8	24.3	53.2	22.6	50.1	21.5	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.8	7.0	-29.4	28.8	4.0	2.2	-32.3	30.3	-34.7	31.6	-37.0	32.9	-39.9	34.5		
(5)			-29.6	20.9	4.0	1.4	-32.5	22.0	-34.8	22.8	-37.1	23.6	-40.0	24.6		

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-8.7	-8.8	-4.1	2.4	9.1	14.5	17.7	15.9	10.5	4.2	-0.9	-5.3
(7)		DBStd	10.43	7.37	7.48	5.09	3.93	4.49	3.70	3.29	3.17	3.27	3.82	4.50	6.81
(8)		HDD10.0	2866	578	526	438	228	72	5	0	1	33	182	327	475
(9)		HDD18.3	5306	837	759	696	478	288	124	51	89	235	438	577	733
(10)		CDD10.0	660	0	0	0	1	44	141	241	183	48	2	0	0
(11)		CDD18.3	57	0	0	0	0	1	10	33	13	0	0	0	0
(12)		CDH23.3	322	0	0	0	0	14	61	187	61	0	0	0	0
(13)		CDH26.7	48	0	0	0	0	1	7	32	8	0	0	0	0
(14)	Wind	WSAvg	3.3	3.1	3.1	3.3	3.0	3.1	3.2	2.9	3.0	3.3	3.6	3.8	3.6
(15)	Precipitation	PrecAvg	655	51	41	35	32	47	70	89	71	53	58	52	53
(16)		PrecMax	800	82	73	69	63	93	131	146	157	102	93	124	86
(17)		PrecMin	536	20	5	7	10	9	17	54	19	14	14	10	20
(18)		PrecStd	79	18	19	13	16	26	31	30	33	26	21	28	19
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.5	4.3	8.6	17.2	25.3	27.8	29.6	27.7	20.6	13.2	9.1	6.5
(20)			MCWB	2.7	2.7	4.3	10.0	14.9	18.0	19.8	18.6	15.4	11.1	8.3	5.8
(21)		2%	DB	2.1	2.6	5.6	13.3	22.1	25.0	27.2	24.9	18.3	11.8	7.4	4.3
(22)			MCWB	1.5	1.8	2.6	7.8	13.5	16.7	19.1	17.6	14.5	10.5	6.7	3.8
(23)		5%	DB	1.1	1.3	3.7	11.0	19.2	23.0	25.4	23.1	16.7	10.7	6.0	3.0
(24)			MCWB	0.8	0.8	1.6	6.0	12.1	15.9	18.2	17.0	13.6	9.6	5.5	2.5
(25)		10%	DB	0.2	0.4	2.2	8.8	16.8	21.0	23.5	21.3	15.2	9.6	4.8	1.9
(26)			MCWB	0.0	0.0	0.7	5.0	10.7	15.0	17.4	16.1	12.8	8.6	4.3	1.4
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	3.1	4.7	11.2	16.4	19.9	21.6	20.2	16.2	11.8	8.4	5.7
(28)			MCDWB	3.2	3.9	7.7	15.3	22.7	24.3	26.4	25.1	18.8	12.7	9.0	6.1
(29)		2%	WB	1.6	1.9	3.0	8.6	14.7	18.4	20.3	18.7	15.2	10.8	6.7	3.8
(30)			MCDWB	1.8	2.4	4.8	12.5	19.9	22.5	24.9	22.8	17.5	11.6	7.2	4.2
(31)		5%	WB	0.8	0.9	1.9	6.7	13.0	17.0	19.3	17.8	14.2	9.7	5.4	2.4
(32)			MCDWB	1.0	1.3	3.2	9.9	17.9	21.3	23.7	21.8	16.3	10.4	5.9	2.8
(33)		10%	WB	0.0	0.1	1.0	5.1	11.4	15.6	18.3	16.9	13.2	8.6	4.2	1.3
(34)			MCDWB	0.3	0.4	2.0	8.0	15.9	20.1	22.3	20.2	15.0	9.4	4.6	1.7
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.6	6.6	7.7	8.2	8.8	7.7	7.5	7.0	5.7	3.9	3.5	4.7
(36)			MCDBR	4.6	5.4	8.0	11.8	12.9	10.4	9.8	9.3	7.4	4.7	4.2	3.8
(37)			MCWBR	4.6	5.1	5.7	7.0	6.1	4.6	4.0	3.7	4.4	3.8	4.1	3.7
(38)		5% WB	MCDBR	4.6	5.3	6.4	11.1	10.9	9.1	8.7	8.2	6.7	4.4	4.1	3.8
(39)			MCWBR	4.6	5.0	4.9	7.0	5.6	4.6	4.1	3.9	4.4	3.9	4.0	3.7
(40)	Clear-Sky Solar Irradiance	taub	0.229	0.286	0.313	0.347	0.345	0.343	0.367	0.350	0.327	0.312	0.246	N/A	
(41)		taud	2.002	2.114	2.184	2.341	2.465	2.494	2.442	2.475	2.528	2.394	2.091	N/A	
(42)		Ebn at Noon	393	636	777	835	873	880	844	828	781	627	369	N/A	
(43)		Edh at Noon	33	67	92	99	95	95	97	86	68	52	29	N/A	
(44)	All-Sky Solar Radiation	RadAvg	0.19	0.74	2.26	3.64	4.78	5.26	5.01	3.82	2.13	0.82	0.20	0.08	
(45)		RadStd	0.03	0.17	0.27	0.44	0.53	0.33	0.51	0.43	0.24	0.14	0.04	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		+0.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (35 neighbors)	+0.57	N/A	N/A	N/A	N/A	N/A	-177	-191	N/A	N/A	(47)

Nomenclature: See separate page